

Relationship Analysis

Revenue Administration Database

1. One-to-One (1:1) Relationships

There are no one-to-one relationships in this database. Each entity can be associated with multiple records in other related tables.

2. One-to-Many (1:M) Relationships

Parent Table	Child Table
TAXPAYER	TAX_REGISTRATION
TAXPAYER	BUSINESS
TAXPAYER	PROPERTY
TAXPAYER	VEHICLE
TAXPAYER	TAX_AUDIT
TAXPAYER	TAX_REFUND
TAXPAYER	TAX_OBJECTION
TAXPAYER	ENFORCEMENT_CASE
TAX_CENTRE	TAX_OFFICER
TAX_CENTRE	TAX_REGISTRATION
TAX_CENTRE	REVENUE_TARGET
TAX_TYPE	TAX_REGISTRATION
TAX_TYPE	TAX_PERIOD
TAX_TYPE	AUDIT_FINDING
TAX_TYPE	REVENUE_TARGET
TAX_REGISTRATION	TAX_DECLARATION
TAX_PERIOD	TAX_DECLARATION
TAX_DECLARATION	TAX_ASSESSMENT
TAX_OFFICER	TAX_ASSESSMENT
TAX_OFFICER	TAX_AUDIT
TAX_OFFICER	ENFORCEMENT_CASE
TAX_AUDIT	AUDIT_FINDING
TAX_ASSESSMENT	TAX_PAYMENT
TAX_ASSESSMENT	PENALTY
TAX_ASSESSMENT	TAX_OBJECTION
BANK	TAX_PAYMENT
TAX_PAYMENT	TAX_REFUND

3. Many-to-Many (M:N) Relationships

There are no direct many-to-many relationships in the final database because they have been resolved using junction tables.

Relationship	Junction Table
TAXPAYER ↔ TAX_TYPE	TAX_REGISTRATION
TAX_CENTRE ↔ TAX_TYPE	REVENUE_TARGET
TAX_REGISTRATION ↔ TAX_PERIOD	TAX_DECLARATION

4. Parent Tables

The following tables are parent tables because their primary keys are referenced by other tables:

- TAXPAYER
- TAX_CENTRE
- TAX_OFFICER
- TAX_TYPE
- BANK
- TAX_REGISTRATION
- TAX_PERIOD
- TAX_DECLARATION
- TAX_ASSESSMENT
- TAX_AUDIT
- TAX_PAYMENT

5. Child Tables

The following tables contain foreign keys that reference parent tables:

- TAX_OFFICER
- TAX_REGISTRATION
- TAX_PERIOD
- TAX_DECLARATION
- TAX_ASSESSMENT
- TAX_PAYMENT
- PENALTY
- TAX_AUDIT
- AUDIT_FINDING
- TAX_REFUND
- TAX_OBJECTION
- ENFORCEMENT_CASE
- BUSINESS
- PROPERTY
- VEHICLE
- REVENUE_TARGET

6. Mandatory Relationships

The following child tables cannot exist without their parent records because they depend on foreign keys:

Child Table	Depends On (Parent Table(s))
TAX_OFFICER	TAX_CENTRE
TAX_REGISTRATION	TAXPAYER, TAX_TYPE, TAX_CENTRE
TAX_PERIOD	TAX_TYPE
TAX_DECLARATION	TAX_REGISTRATION, TAX_PERIOD
TAX_ASSESSMENT	TAX_DECLARATION, TAX_OFFICER
TAX_PAYMENT	TAX_ASSESSMENT, BANK
PENALTY	TAX_ASSESSMENT
TAX_AUDIT	TAXPAYER, TAX_OFFICER
AUDIT_FINDING	TAX_AUDIT, TAX_TYPE
TAX_REFUND	TAXPAYER, TAX_PAYMENT
TAX_OBJECTION	TAXPAYER, TAX_ASSESSMENT
ENFORCEMENT_CASE	TAXPAYER, TAX_OFFICER
BUSINESS	TAXPAYER
PROPERTY	TAXPAYER
VEHICLE	TAXPAYER
REVENUE_TARGET	TAX_CENTRE, TAX_TYPE

7. Optional Relationships

A parent record may exist without related child records. For example:

- A taxpayer may have no business.
- A taxpayer may have no property.
- A taxpayer may have no vehicle.
- A taxpayer may never be audited.
- A taxpayer may never request a refund.
- A taxpayer may never file an objection.
- A tax officer may never conduct an audit.
- A bank may not receive any tax payments.

- A tax centre may not yet have a revenue target.

8. Primary Key and Foreign Key Relationships

Parent Primary Key	Child Foreign Key
taxpayer_id	taxpayer_id
tax_centre_id	tax_centre_id
officer_id	officer_id
tax_type_id	tax_type_id
registration_id	registration_id
tax_period_id	tax_period_id
declaration_id	declaration_id
assessment_id	assessment_id
audit_id	audit_id
payment_id	payment_id
bank_id	bank_id

9. Referential Integrity Rules

- Every foreign key must reference an existing primary key.
- A child record cannot exist without its parent record.
- Primary keys must be unique.
- Parent records should not be deleted while child records reference them unless cascading actions are defined.
- Updates to primary keys must maintain foreign key consistency.
- Foreign key values must always match existing parent records.

10. Correct Order for Creating Tables

- | | |
|---------------------|----------------------|
| 1. TAXPAYER | 11. REVENUE_TARGET |
| 2. TAX_CENTRE | 12. TAX_DECLARATION |
| 3. TAX_TYPE | 13. TAX_AUDIT |
| 4. BANK | 14. TAX_ASSESSMENT |
| 5. TAX_OFFICER | 15. AUDIT_FINDING |
| 6. TAX_REGISTRATION | 16. TAX_PAYMENT |
| 7. TAX_PERIOD | 17. PENALTY |
| 8. BUSINESS | 18. TAX_REFUND |
| 9. PROPERTY | 19. TAX_OBJECTION |
| 10. VEHICLE | 20. ENFORCEMENT_CASE |

11. Correct Order for Inserting Data

Data should be inserted in the same dependency order used when creating the tables:

- | | |
|---------------------|----------------------|
| 1. TAXPAYER | 11. REVENUE_TARGET |
| 2. TAX_CENTRE | 12. TAX_DECLARATION |
| 3. TAX_TYPE | 13. TAX_AUDIT |
| 4. BANK | 14. TAX_ASSESSMENT |
| 5. TAX_OFFICER | 15. AUDIT_FINDING |
| 6. TAX_REGISTRATION | 16. TAX_PAYMENT |
| 7. TAX_PERIOD | 17. PENALTY |
| 8. BUSINESS | 18. TAX_REFUND |
| 9. PROPERTY | 19. TAX_OBJECTION |
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